

Exam 2 Questions (5.C – 8.C)

CHEM 1211K | Fall 2025

This exam includes coverage of solution stoichiometry, fundamentals of energy, heat, and work, the first law of thermodynamics, calorimetry, fundamentals of enthalpy, calculations of enthalpy involving Hess's law, gas laws, the kinetic theory of gases, the second law of thermodynamics, entropy, free energy, classical and quantum models of light, and the Bohr model of the hydrogen atom.

QUESTION 1

1. A solution with a total volume of 150. mL is prepared by dissolving 3.00 g borneol ($\text{C}_{10}\text{H}_{18}\text{O}$, 154.3 g/mol) in a liquid mixture of water and acetonitrile. What is the molarity of borneol in this solution?

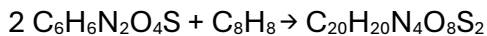
A. ☐ 0.130 M
B. ☐ 0.0195 M
C. ☐ 20.0 M
D. ☐ 0.0200 M
E. ☐ 0.271 M
2. A solution with a total volume of 20.0 mL is prepared by dissolving 3.30 g bromocyclohexane ($\text{C}_6\text{H}_{11}\text{Br}$, 163.1 g/mol) in liquid toluene. What is the molarity of bromocyclohexane in this solution?

A. ☐ 1.01 M
B. ☐ 0.165 M
C. ☐ 6.06 M
D. ☐ 0.282 M
E. ☐ 0.491 M
3. A solution with a total volume of 20.0 mL is prepared by dissolving 8.50 g nosylamide ($\text{C}_6\text{H}_6\text{N}_2\text{O}_4\text{S}$, 202.2 g/mol) in liquid acetonitrile. What is the molarity of nosylamide in this solution?

A. ☐ 0.425 M
B. ☐ 2.35 M
C. ☐ 0.663 M
D. ☐ 2.10 M
E. ☐ 3.15 M

QUESTION 2

4. Nosylamide (**A**, $\text{C}_6\text{H}_6\text{N}_2\text{O}_4\text{S}$, 202.2 g/mol) reacts with styrene (**B**, C_8H_8 , 104.2 g/mol) in solution according to the balanced chemical equation below.



In one run of this reaction, 84.0 mL of a 0.500 M solution of nosylamide were combined with 100. mL of a 0.200 M solution of styrene. Which reactant is limiting in this reaction? [☐ Styrene ☐ Nosylamide ☐ Neither]

What is the theoretical *molarity* of the product in the solution at the end of the reaction? Assume no change in solution volume during the reaction.

[☐ 0.109 M ☐ 0.238 M ☐ 0.228 M]

QUESTION 3

5. Which of the following solutions has the greatest *total* concentration of dissolved ions?

- A. ☐ 0.70 M KCl
- B. ☐ 0.50 M Na_2CO_3
- C. ☐ 0.50 M LiBr
- D. ☐ 0.25 M K_3PO_4
- E. ☐ 0.50 M KClO_4

6. Which of the following solutions has the smallest *total* concentration of dissolved ions?

- A. ☐ 0.50 M Na_2CO_3
- B. ☐ 0.75 M KCl
- C. ☐ 0.50 M MgBr_2
- D. ☐ 0.25 M K_3PO_4
- E. ☐ 0.50 M K_2SO_4

7. Which of the following solutions has the greatest *total* concentration of dissolved ions? Assume a volume of 1.00 L for all solutions.

- A. ☐ 0.50 M Na_2CO_3
- B. ☐ 0.75 M KCl
- C. ☐ 0.50 M K_3PO_4
- D. ☐ 0.50 M LiBr
- E. ☐ 0.25 M KClO_4

QUESTION 4

8. Indicate whether each statement below is true or false.

[S1] Heat flow from a colder body to a hotter body is a violation of the first law of thermodynamics.

[S2] Heat, work, and energy all share the same units.

[S3] The specific heat of liquid water is equal to the specific heat of solid water (ice).

[S4] In any process in which a system does work on its surroundings, the internal energy of the system must decrease.

9. Indicate whether each statement below is true or false.

[S1] Heat flow from a colder body to a hotter body is a violation of the second law of thermodynamics.

[S2] Heat, work, and energy all share the same units.

[S3] The specific heat of liquid water is equal to the specific heat of solid water (ice).

[S4] When a system does work on its surroundings, the internal energy of the system may remain unchanged.

10. Indicate whether each statement below is true or false.

[S1] Heat flow from a colder body to a hotter body is a violation of the second law of thermodynamics.

[S2] Heat, work, and energy have different units.

[S3] The specific heat of liquid water is equal to the specific heat of solid water (ice).

[S4] When a system does work on its surroundings, the internal energy of the system may remain unchanged.

QUESTION 5

11. For which of the following processes is the measured heat q *NOT* equal to the enthalpy change of the system ΔH ?

- A. ☐ In a closed and rigid vessel, hydrogen gas reacts with ethylene gas to form ethane gas: $\text{H}_2(\text{g}) + \text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$.
- B. ☐ A gas expands against a constant external pressure with an increase in number of moles that keeps the pressure of the gas constant.
- C. ☐ In an open flask, liquid ethanol evaporates to form a gas.
- D. ☐ For **all** these processes, $q = \Delta H$.
- E. ☐ For **none** of these processes, $q = \Delta H$.

12. For which of the following processes is the measured heat q equal to the enthalpy change of the system ΔH ?

- A. ☐ In a flexible balloon, hydrogen gas reacts with ethylene gas to form ethane gas at constant pressure: $\text{H}_2(\text{g}) + \text{C}_2\text{H}_4(\text{g}) \rightarrow \text{C}_2\text{H}_6(\text{g})$.
- B. ☐ A gas expands against a constant external pressure with an increase in number of moles that keeps the pressure of the gas constant.
- C. ☐ In an open flask, liquid ethanol evaporates to form a gas.
- D. ☐ For all these processes, $q = \Delta H$.
- E. ☐ For **none** of these processes, $q = \Delta H$.

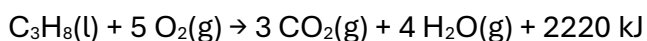
QUESTION 6

13. An electric car battery requires 2.0×10^6 Joules of energy to fully recharge. Consider a setup in which the combustion of octane (C_8H_{18}) is used to supply this energy. Assuming the energy transfer occurs without any losses, how many moles of octane must be combusted to supply this amount of energy?

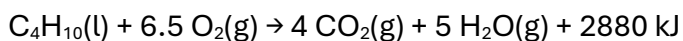


- A. ☐ 370 mol
- B. ☐ 2.7 mol
- C. ☐ 0.0061 mol
- D. ☐ 0.37 mol
- E. ☐ 1.2 mol

14. An electric car battery requires 1.5×10^6 Joules of energy to fully recharge. Consider a setup in which the combustion of propane (C_3H_8) is used to supply this energy. Assuming the energy transfer occurs without any losses, how many moles of propane must be combusted to supply this amount of energy?



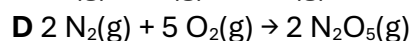
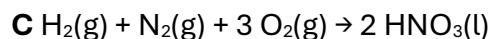
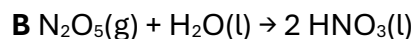
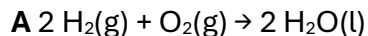
- A. ☐ 680 mol
 - B. ☐ 1.5 mol
 - C. ☐ 2.7 mol
 - D. ☐ 0.083 mol
 - E. ☐ 0.68 mol
15. An electric car battery requires 1.5×10^6 Joules of energy to fully recharge. Consider a setup in which the combustion of butane (C_4H_{10}) is used to supply this energy. Assuming the energy transfer occurs without any losses, how many moles of butane must be combusted to supply this amount of energy?



- A. ☐ 0.52 mol
- B. ☐ 1.9 mol
- C. ☐ 520 mol
- D. ☐ 0.0061 mol
- E. ☐ 2.7 mol

QUESTION 7

16. Which of the following equations shows how reactions **A – C** can be combined to generate reaction **D**?



A. ☐ **D = -A - 2B + 2C**

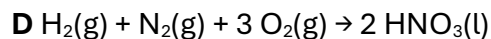
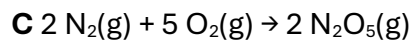
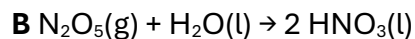
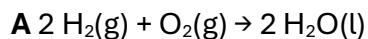
B. ☐ **D = A + 2B - 2C**

C. ☐ **D = A + B - 2C**

D. ☐ **D = -2A - B + 2C**

E. ☐ **D = A - B - C**

17. Which of the following equations shows how reactions **B – D** can be combined to generate reaction **A**?



A. ☐ **A = -2B + C + D**

B. ☐ **A = -2B - C + 2D**

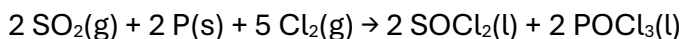
C. ☐ **A = C - B + 2D**

D. ☐ **A = -2B + C - D**

E. ☐ **A = B - C - D**

QUESTION 8

18. Use the standard enthalpies of formation given below to determine $\Delta H^\circ_{\text{rxn}}$ for the indicated reaction. Enter your response in kilojoules (kJ) to the nearest 1 kJ with appropriate sign.

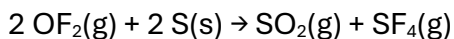


$\text{SO}_2(\text{g})$ -297 kJ

$\text{SOCl}_2(\text{l})$ -313 kJ

$\text{POCl}_3(\text{l})$ -598 kJ

19. Use the standard enthalpies of formation given below to determine $\Delta H^\circ_{\text{rxn}}$ for the indicated reaction. Enter your response in kilojoules (kJ) to the nearest 1 kJ with appropriate sign.



$\text{OF}_2(\text{g})$ $+25 \text{ kJ}$

$\text{SO}_2(\text{g})$ -297 kJ

$\text{SF}_4(\text{g})$ -763 kJ

QUESTION 9

20. When 0.187 g of benzene (C_6H_6) is burned in a bomb calorimeter, the temperatures of the water bath and calorimeter increase by 4.53°C . The water bath has a mass of 250. g. What heat capacity of the calorimeter is implied by these results?

The specific heat of water is $4.18 \text{ J g}^{-1} ^\circ\text{C}^{-1}$ and the specific enthalpy of combustion of benzene is -41.9 kJ g^{-1} .

- A. ☐ $1040 \text{ J } ^\circ\text{C}^{-1}$
- B. ☐ $685 \text{ J } ^\circ\text{C}^{-1}$
- C. ☐ $1730 \text{ J } ^\circ\text{C}^{-1}$
- D. ☐ $894 \text{ J } ^\circ\text{C}^{-1}$
- E. ☐ $412 \text{ J } ^\circ\text{C}^{-1}$

21. When 0.280 g of benzene (C_6H_6) is burned in a bomb calorimeter, the temperatures of the water bath and calorimeter increase by 6.80°C . The water bath has a mass of 250. g. What heat capacity of the calorimeter is implied by these results?

The specific heat of water is $4.18 \text{ J g}^{-1} ^\circ\text{C}^{-1}$ and the specific enthalpy of combustion of benzene is -41.9 kJ g^{-1} .

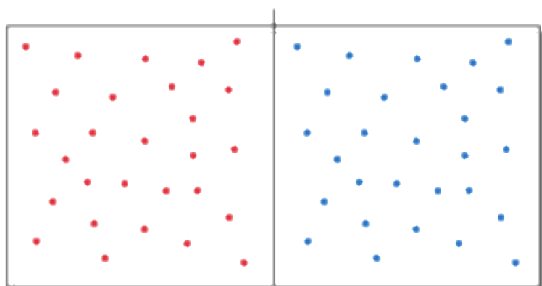
- A. ☐ $1040 \text{ J } ^\circ\text{C}^{-1}$
 - B. ☐ $1730 \text{ J } ^\circ\text{C}^{-1}$
 - C. ☐ $894 \text{ J } ^\circ\text{C}^{-1}$
 - D. ☐ $686 \text{ J } ^\circ\text{C}^{-1}$
 - E. ☐ $412 \text{ J } ^\circ\text{C}^{-1}$
22. When 0.374 g of benzene (C_6H_6) is burned in a bomb calorimeter, the temperatures of the water bath and calorimeter increase by 9.06°C . The water bath has a mass of 250. g. What heat capacity of the calorimeter is implied by these results?

The specific heat of water is $4.18 \text{ J g}^{-1} ^\circ\text{C}^{-1}$ and the specific enthalpy of combustion of benzene is -41.9 kJ g^{-1} .

- A. ☐ $1040 \text{ J } ^\circ\text{C}^{-1}$
- B. ☐ $685 \text{ J } ^\circ\text{C}^{-1}$
- C. ☐ $1730 \text{ J } ^\circ\text{C}^{-1}$
- D. ☐ $894 \text{ J } ^\circ\text{C}^{-1}$
- E. ☐ $412 \text{ J } ^\circ\text{C}^{-1}$

QUESTION 10

23. Red and blue ideal gases are present in two compartments of equal volume separated by a partition at constant temperature. Consider the process that occurs when the partition is removed and predict the end state at equilibrium (when no further macroscopic change is observed). Indicate the signs of ΔH , ΔS , and ΔG for a system consisting of the two gases.



ΔH is [☐ positive ☐ negative ☐ zero].

ΔS is [☐ positive ☐ negative ☐ zero].

ΔG is [☐ positive ☐ negative ☐ zero].

QUESTION 11

24. Indicate whether the sign of ΔS for each process below is positive or negative.

$\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ [☐ positive ☐ negative]

$2 \text{H}_3\text{O}^+(\text{aq}) + \text{CO}_3^{2-}(\text{aq}) \rightarrow \text{CO}_2(\text{g}) + 3 \text{H}_2\text{O}(\text{l})$ [☐ positive ☐ negative]

Freezing of water at 1°C [☐ positive ☐ negative]

25. Indicate whether the sign of ΔS for each process below is positive or negative.

$\text{PCl}_5(\text{g}) \rightarrow \text{PCl}_3(\text{g}) + \text{Cl}_2(\text{g})$ [☐ positive ☐ negative]

$\text{Zn}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow \text{ZnCl}_2(\text{s})$ [☐ positive ☐ negative]

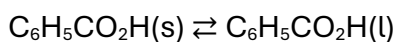
Melting of ice at -1°C [☐ positive ☐ negative]

26. Indicate whether the sign of ΔS for each process below is positive or negative.



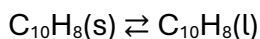
QUESTION 12

27. The melting point of benzoic acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$) is the temperature at which $\Delta G = 0$ for the following process.



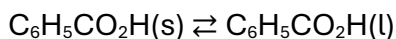
Determine the melting point of benzoic acid in degrees Celsius ($^\circ\text{C}$) given ΔH for melting is $+18.0\text{ kJ mol}^{-1}$ and ΔS for melting is $+45.6\text{ J K}^{-1}\text{ mol}^{-1}$. Enter your response to the nearest $1\text{ }^\circ\text{C}$.

28. The melting point of naphthalene (C_{10}H_8) is the temperature at which $\Delta G = 0$ for the following process.



The melting point of naphthalene is $128\text{ }^\circ\text{C}$ and ΔS for melting is $+47.7\text{ J K}^{-1}\text{ mol}^{-1}$. Determine ΔH for melting of naphthalene. Enter your response in kilojoules per mole (kJ mol^{-1}) to the nearest 0.1 kJ mol^{-1} and with appropriate sign.

29. The melting point of benzoic acid ($\text{C}_6\text{H}_5\text{CO}_2\text{H}$) is the temperature at which $\Delta G = 0$ for the following process.



The melting point of benzoic acid is $122\text{ }^\circ\text{C}$ and ΔH for melting is $+18.0\text{ kJ mol}^{-1}$. Determine ΔS for melting of benzoic acid. Enter your response in Joules per Kelvin per mole ($\text{J K}^{-1}\text{ mol}^{-1}$) to the nearest $1\text{ J K}^{-1}\text{ mol}^{-1}$ and with appropriate sign.

QUESTION 13

30. Ethylene gas initially at a pressure of 1.38 atm, a volume of 2.00 L, and a temperature of 305 K is compressed to a volume of 0.750 L and pressure 3.00 atm. What is the temperature of the gas after this compression?

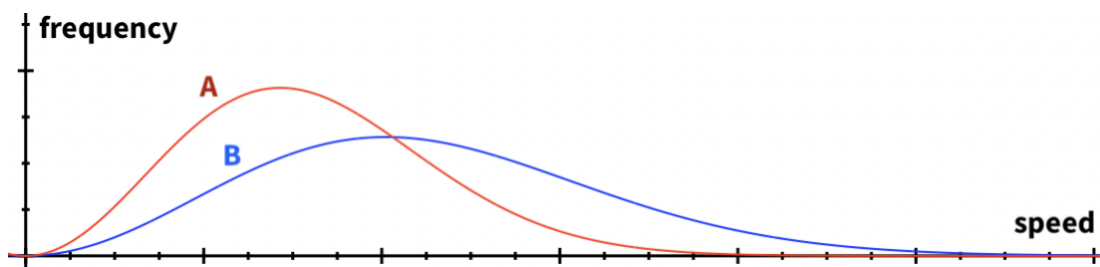
Enter your response in Kelvin (K) to the nearest 1 K.

31. Ethylene gas initially at a pressure of 3.00 atm, a volume of 0.750 L, and a temperature of 249 K expands to a volume of 2.00 L and pressure 1.38 atm. What is the temperature of the gas after this expansion?

Enter your response in Kelvin (K) to the nearest 1 K.

QUESTION 14

32. The image below shows speed distributions for two ideal gases **A** and **B** at thermal equilibrium in a single container. Fill in the blanks below to describe these gases.

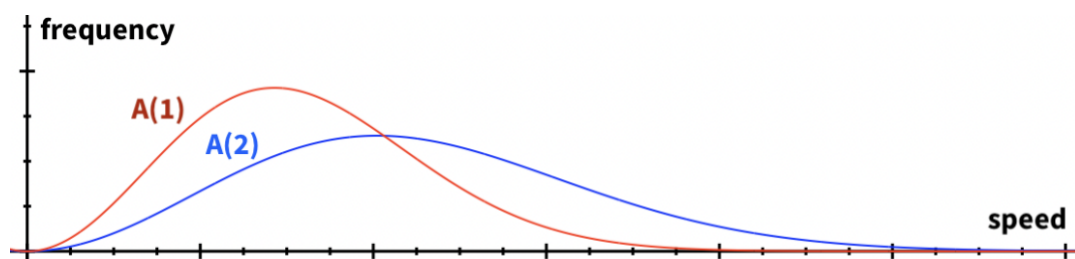


A and B have [☐ identical ☐ different] molar masses.

A and B have [☐ identical ☐ different] mean kinetic energies.

For both A and B, the most probable speed is [☐ less than ☐ equal to ☐ greater than] the root mean square speed.

33. The image below shows speed distributions for two samples of an ideal gas **A** with equal numbers of moles at two temperatures T_1 and T_2 . Fill in the blanks below to describe these gases.



T_1 is [☐ greater than ☐ equal to ☐ less than] T_2 .

The average kinetic energy of gas A at T_1 is [☐ greater than ☐ equal to ☐ less than] the average kinetic energy of gas A at T_2 .

For both A(1) and A(2), the root mean square speed is [☐ less than ☐ equal to ☐ greater than] the most probable speed.

QUESTION 15

34. Consider an apparatus designed to measure the photoelectric effect with a lamp shining on a piece of metal and circuits to measure the current and kinetic energy of any ejected electrons. Which of the following interventions would cause the measured current to increase?

- A. ☐ Increasing the intensity of light with wavelength *shorter* than the threshold wavelength
- B. ☐ Increasing the intensity of light with wavelength *longer* than the threshold wavelength
- C. ☐ Increasing the wavelength of light with wavelength *longer* than the threshold wavelength
- D. ☐ Increasing the wavelength of light with wavelength *shorter* than the threshold wavelength
- E. ☐ None of these

35. Consider an apparatus designed to measure the photoelectric effect with a lamp shining on a piece of metal and circuits to measure the current and kinetic energy of any ejected electrons. Which of the following interventions would cause the measured kinetic energy to increase?

- A. ☐ Increasing the frequency of light with frequency *greater* than the threshold frequency
- B. ☐ Increasing the intensity of light with wavelength *shorter* than the threshold wavelength
- C. ☐ Increasing the intensity of light with wavelength *longer* than the threshold wavelength
- D. ☐ Decreasing the frequency of light with frequency *smaller* than the threshold frequency
- E. ☐ None of these

QUESTION 16

36. An observed line in the atomic *emission* spectrum of hydrogen has a wavelength of 103 nm. Use the dropdown menus below to indicate the initial and final energy levels for the electronic transition associated with this line. The Rydberg constant is 0.01097 nm^{-1} .

The initial energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.

The final energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.

37. An observed line in the atomic *emission* spectrum of hydrogen has a wavelength of 97.3 nm. Use the dropdown menus below to indicate the initial and final energy levels for the electronic transition associated with this line. The Rydberg constant is 0.01097 nm^{-1} .

The initial energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.

The final energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.

38. An observed line in the atomic *emission* spectrum of hydrogen has a wavelength of 486.1 nm. Use the dropdown menus below to indicate the initial and final energy levels for the electronic transition associated with this line. The Rydberg constant is 0.01097 nm^{-1} .

The initial energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.

The final energy level is $n = [\text{ } \text{ } 1 \text{ } \text{ } 2 \text{ } \text{ } 3 \text{ } \text{ } 4 \text{ } \text{ } 5 \text{ }]$.